

Profile

An adaptable person with a PhD in the physical sciences and several years spent as a researcher in early-stage drug discovery. Along with skills as a research software engineer and data scientist, I have specialist knowledge of machine vision tasks, including creating deep learning workflows to aid understanding of large 3-dimensional datasets as well as training and deploying models for image classification and object detection.

Relevant Work History

- March 2024 – Present* **Senior Research Software Engineer (with focus on ML and AI). (Full time). University of Oxford., UK**
Responsible for supporting postdoctoral research fellows on the Eric & Wendy Schmidt AI in Science Programme. Helping fellows to achieve goals in their projects as well as teaching software engineering best practices.
Example projects: Tracking animals in video data, creating and training models for fish pose classification, using deep features for individual animal identification, deep learning workflows on High Performance Computing infrastructure, troubleshooting deep reinforcement learning for inverse design of nanophotonic devices, visualising the principal components of bird flight
- May 2023 – March 2024* **Application Scientist (Full time). Dotmatics Ltd., UK**
Customer-facing role where I configured the Dotmatics laboratory informatics and electronic notebook systems for bespoke science workflows, mainly for clients in the pharmaceutical sector. Tasks included data extraction, transformation and loading using SQL, integration of data from legacy systems, creation of customer training materials, working with clients in agile project processes, pre-sales estimation of project time and resources, and troubleshooting technical issues.
- Dec 2018 – Apr 2023* **Data Analysis Scientist – Machine Vision. (Full time). Diamond Light Source Ltd., UK**
Several projects including: (i) extending SuRVoS (Super-Region Volume Segmentation) Workbench software, which allows interactive machine learning segmentation of 3D image data; (ii) creating an automated deep learning package for segmentation of multi-gigabyte image volumes; and (iii) creating deep learning pipelines to classify thousands of micrographs and to find locations of crystals for automated data collection (*Recognition of Outstanding Contribution Award*, April 2023). **Other responsibilities:** Line management of graduate software engineers. Recruitment and supervision of Year in Industry students. Member of selection and interview panels for research software engineer positions. I ran a series of regular meetings for staff working on image analysis and acted as a point of contact for newer team members.
- Sept 2017 – Dec 2018* **Software Engineer. (Full time). Diamond Light Source Ltd., UK**
Maintained and developed features for Generic Data Acquisition (GDA) software – a large Java Eclipse RCP application. I supported data collection on five beamline branches (experimental stations) that study the electronic surface properties of materials using Photoelectron Spectroscopy (PES). Participated in agile sprints to develop core software features
- Jan 2017 – Aug 2017* **Software Engineer. (Full time). Zoetrope Ltd., UK**
Created RESTful APIs in Python backed by MongoDB. I also wrote an in-depth research report for a client in a consulting capacity and implemented an end-to-end testing setup that integrated serial communications to hardware with server-side software and a mobile app.

Relevant Work History (Continued)

University of Oxford, Oxford, UK

- 2009 - 2014** **Postdoctoral Research Assistant** (Full time, multiple contracts), Department of Chemistry
- I worked on two main projects: The first was a collaboration that included GSK, Pfizer, and the US National Institutes of Health. This aimed to develop early-stage drugs (chemical probes) to aid understanding of enzymes involved in epigenetic regulation and implicated in developmental disorders and cancers. The second project, funded by the British Heart Foundation, aimed to develop cardiac treatments using novel peptide inhibitors.
- For both projects, I carried out independent research using a wide range of techniques, including activity assays, crystallography and mass spectrometry, I developed new methodologies and analysed large quantities of experimental data. I co-supervised a master's student and gave training and advice to many others in this busy lab environment with a high proportion of student scientists.
- 2006 - 2009** **Research Assistant** (Full time), Structural Genomics Consortium
- Member of several teams across a high-throughput environment that generates open-source protein structural information for the science community. I purified, crystallised, and performed biophysical and biochemical analyses on a wide range of human proteins related to disease as well as managing compound libraries for screening. I also co-supervised an undergraduate summer student and a master's student.

Education

MSc. Computer Science. University of Bristol, Department of Computer Science. *2015 - 2016.*

Distinction (78%)

Courses taken included: Programming in C and Java, Computer Architecture, Web Technologies and Databases.

My research project used texture analysis in combination with machine learning (clustering, random forests) to detect crystals in microscope images. Work was done in collaboration with the Structural Genomics Consortium, University of Oxford and iXpressGenes Ltd., Huntsville, Alabama.

D.Phil. Materials Science. University of Oxford, Department of Materials. *2001 - 2005.*

Awarded funding by the Engineering and Physical Sciences Research Council for an independent research project to develop novel, improved coatings for orthopaedic implants with drug-delivery capability. Analysis techniques included scanning and transmission electron microscopy, electron diffraction and X-ray powder diffraction crystallography.

Chair of the departmental Joint Consultative Committee for Graduates. Social Secretary of St. Catherine's College middle common room and member of the college IT committee.

BSc. (Hons.) Biology. University of Nottingham, School of Life and Environmental Sciences. *1997 - 2000.*

Upper Second Class (69%)

Broad course with topics ranging from Cancer Biology to Primatology. Final year research project on the physiological effects of a gene insertion designed to increase the shelf life of plants. Reached the national final of the Biotechnology Young Entrepreneurs Scheme competition where our team pitched a business plan for a surgical adhesive product.

Programming Experience

Python (NumPy, PyTorch, Pandas), SQL. Also experience of: Java, C, MATLAB, JavaScript, HTML, CSS. I use Git for version control and have set up continuous integration pipelines on both GitLab and Github. I have used Bash scripting for Linux along with SGE for submitting jobs to High Performance Computing clusters.

Interests

I love to travel and attend gigs, otherwise, I enjoy photography, running and yoga. I also take a keen interest in science and technology developments, especially those with environmental and societal benefits.

Publications

From Voxels to Viruses: Using Deep Learning and Crowdsourcing to Understand a Virus Factory. Pennington, A., King, O.N.F., Tun, W.M., Boyce, M., Sutton, G., Stuart, D.I., Basham, M and Darrow, M.C. *Citizen Science: Theory and Practice*, 2024. **9** (1), 37. DOI: [10.5334/cstp.739](https://doi.org/10.5334/cstp.739)

CHiMP: deep-learning tools trained on protein crystallization micrographs to enable automation of experiments. King, O.N.F., Levik, K.E., Sandy, J. and Basham, M. *Acta Crystallographica Section D*, 2024. **80** (10), 744-764. DOI: [10.1107/S2059798324009276](https://doi.org/10.1107/S2059798324009276)

Online citizen science with the Zooniverse for analysis of biological volumetric data.

Smith, P., King, O.N.F., Pennington, A., Tun, W.M., Basham, M., Jones, M.L., Collinson, L.M., Darrow, M. C., Spiers, H. *Histochemistry and Cell Biology*, 2023. **160**, 253–276, DOI: [10.1007/s00418-023-02204-6](https://doi.org/10.1007/s00418-023-02204-6)

Comparison of Methods to Segment Variable-Contrast XCT Images of Methane-Bearing Sand Using U-Nets Trained on Single Dataset Sub-Volumes.

Alvarez-Borges FJ, King ONF, Madhusudhan BN, Connolley T, Basham M, Ahmed SI. *Methane*, 2023, (2)1–23. DOI: [10.3390/methane2010001](https://doi.org/10.3390/methane2010001)

Volume Segmantics: A Python Package for Semantic Segmentation of Volumetric Data Using Pre-trained PyTorch Deep Learning Models.

King ONF, Bellos, D and Basham M. *Journal of Open Source Software*. 2022. **7**(78), 4691, DOI: [10.21105/joss.04691](https://doi.org/10.21105/joss.04691)

SuRVoS 2: Accelerating Annotation and Segmentation for Large Volumetric Bioimage Workflows Across Modalities and Scales.

Pennington, A., King, O.N.F., Tun, W.M., Ho, E.M.L., Luengo, I., Darrow, M.C., and Basham, M. *Front Cell Dev Bio*. 2022. DOI: [10.3389/fcell.2022.842342](https://doi.org/10.3389/fcell.2022.842342).

Beamline K11 DIAD: A new instrument for dual imaging and diffraction at Diamond Light Source.

Reinhard C, Drakopoulos M, Ahmed SI, Deyhle H, James A, Charlesworth CM, Burt M, Sutter J, Alexander S, Garland P, Yates T, Marshall R, Kemp B, Warrick E, Pueyos A, Bradnick B, Nagni M, Winter AD, Filik J, Basham M, Wadeson N, King ONF, Aslani N, Dent AJ. 2021. *J Synch Rad*. DOI:[10.1107/S1600577521009875](https://doi.org/10.1107/S1600577521009875)

A massively multi-scale approach to characterising tissue architecture by synchrotron micro-CT applied to the human placenta.

Tun WM, Poologasundarampillai G, Bischof H, Nye G, King ONF, Basham M, Tokudome Y, Lewis RM, Johnstone ED, Brownbill P, Darrow M, Chernyavsky IL. *J R Soc Interface*. 2021. **18**(179):20210140, DOI: [10.1098/rsif.2021.0140](https://doi.org/10.1098/rsif.2021.0140)

Non-competitive cyclic peptides for targeting enzyme-substrate complexes.

McAllister TE, Yeh TL, Abboud MI, Leung IKH, Hookway ES, King ONF, Bhushan B, Williams ST, Hopkinson RJ, Münzel M, Loik ND, Chowdhury R, Oppermann U, Claridge TDW, Goto Y, Suga H, Schofield CJ, Kawamura A. *Chem Sci*. 2018 **9**(20):4569-4578, DOI: [10.1039/c8sc00286j](https://doi.org/10.1039/c8sc00286j)

Highly selective inhibition of histone demethylases by de novo macrocyclic peptides.

Kawamura A, Münzel M, Kojima T, Yapp C, Bhushan B, Goto Y, Tumber A, Katoh T, King ON, Passioura T, Walport LJ, Hatch SB, Madden S, Müller S, Brennan PE, Chowdhury R, Hopkinson RJ, Suga H, Schofield CJ. *Nat Commun*. 2017 **Apr 6**; 8:14773, DOI: [10.1038/ncomms14773](https://doi.org/10.1038/ncomms14773)

Pan-histone demethylase inhibitors simultaneously targeting Jumonji C and lysine-specific demethylases display high anticancer activities.

Rotili D, Tomassi S, Conte M, Benedetti R, Tortorici M, Ciossani G, Valente S, Marrocco B, Labella D, Novellino E, Mattevi A, Altucci L, Tumber A, Yapp C, King ON, Hopkinson RJ, Kawamura A, Schofield CJ, Mai A. *J Med Chem*. 2014 **57**(1):42-55, DOI: [10.1021/jm4012802](https://doi.org/10.1021/jm4012802)

5-Carboxy-8-hydroxyquinoline is a broad spectrum 2-oxoglutarate oxygenase inhibitor which causes iron translocation.

Hopkinson RJ, Tumber A, Yapp C, Chowdhury R, Aik WS, Che KH, Li XS, Kristensen JBL, King ONF, Chan MC, Yeoh KK, Choi H, Walport LJ, Thinnies CC, Bush JT, Lejeune C, Rydzik AM, Rose NR, Bagg EA, McDonough MA, Krojer, TJ, Yue WW, Ng SS, Olsen L, Brennan PE, Oppermann U, Muller S, Klose RJ, Ratcliffe PJ, Schofield CJ, Kawamura A. *Chem Sci*. 2013; **4**:3110-3117, DOI: [10.1039/C3SC51122G](https://doi.org/10.1039/C3SC51122G)

Discovery of ML324, a JMJD2 demethylase inhibitor with demonstrated antiviral activity.

Rai G, Kawamura A, Tumber A, Liang Y, Vogel JL, Arbuckle JH, Rose NR, Dexheimer TS, Foley TL, King ON, Quinn A, Mott BT, Schofield CJ, Oppermann U, Jadhav A, Simeonov A, Kristie TM, Maloney DJ. *Probe Reports from the NIH Molecular Libraries Program*. 2010-2012 Dec 17. <https://www.ncbi.nlm.nih.gov/books/NBK169450/>

Plant growth regulator daminozide is a selective inhibitor of human KDM2/7 histone demethylases.

Rose NR, Woon EC, Tumber A, Walport LJ, Chowdhury R, Li XS, King ON, Lejeune C, Ng SS, Krojer T, Chan MC, Rydzik AM, Hopkinson RJ, Che KH, Daniel M, Strain-Damerell C, Gileadi C, Kochan G, Leung IK, Dunford J, Yeoh KK, Ratcliffe PJ, Burgess-Brown N, von Delft F, Muller S, Marsden B, Brennan PE, McDonough MA, Oppermann U, Klose RJ, Schofield CJ, Kawamura A. *J Med Chem.* 2012; **55**(14):6639-43, DOI: [10.1021/jm300677j](https://doi.org/10.1021/jm300677j)

The *Cryptosporidium parvum* kinome.

Artz JD, Wernimont AK, Allali-Hassani A, Zhao Y, Amani M, Lin YH, Senisterra G, Wasney GA, Fedorov O, King O, Roos A, Lunin VV, Qiu W, Finerty P Jr, Hutchinson A, Chau I, von Delft F, MacKenzie F, Lew J, Kozieradzki I, Vedadi M, Schapira M, Zhang C, Shokat K, Heightman T, Hui R. *BMC Genomics.* 2011; **12**:478, DOI: [10.1186/1471-2164-12-478](https://doi.org/10.1186/1471-2164-12-478)

Inhibition of 2-oxoglutarate dependent oxygenases.

Rose NR, McDonough MA, King ON, Kawamura A, Schofield CJ.

Chem Soc Rev. 2011; **40**(8):4364-97, DOI: [10.1039/c0cs00203h](https://doi.org/10.1039/c0cs00203h)

The oncometabolite 2-hydroxyglutarate inhibits histone lysine demethylases.

Chowdhury R, Yeoh KK, Tian YM, Hillringhaus L, Bagg EA, Rose NR, Leung IK, Li XS, Woon EC, Yang M, McDonough MA, King ON, Clifton IJ, Klose RJ, Claridge TD, Ratcliffe PJ, Schofield CJ, Kawamura A. *EMBO Rep.* 2011; **12**(5):463-9, DOI: [10.1038/embor.2011.43](https://doi.org/10.1038/embor.2011.43)

Inhibition of histone demethylases by 4-carboxy-2,2'-bipyridyl compounds.

Chang KH*, King ON*, Tumber A*, Woon EC, Heightman TD, McDonough MA, Schofield CJ, Rose NR. *ChemMedChem.* 2011; **6**(5):759-64, DOI: [10.1002/cmdc.201100026](https://doi.org/10.1002/cmdc.201100026)

A prominent β -hairpin structure in the winged-helix domain of RECQ1 is required for DNA unwinding and oligomer formation.

Lucic B, Zhang Y, King O, Mendoza-Maldonado R, Berti M, Niesen FH, Burgess-Brown NA, Pike AC, Cooper CD, Gileadi O, Vindigni A. *Nucleic Acids Res.* 2011; **39**(5):1703-17, DOI: [10.1093/nar/gkq1031](https://doi.org/10.1093/nar/gkq1031)

Specific CLK inhibitors from a novel chemotype for regulation of alternative splicing.

Fedorov O, Huber K, Eisenreich A, Filippakopoulos P, King O, Bullock AN, Szklarczyk D, Jensen LJ, Fabbro D, Trappe J, Rauch U, Bracher F, Knapp S. *Chem Biol.* 2011; **18**(1):67-76, DOI: [10.1016/j.chembiol.2010.11.009](https://doi.org/10.1016/j.chembiol.2010.11.009)

Quantitative high-throughput screening identifies 8-hydroxyquinolines as cell-active histone demethylase inhibitors.

King ON*, Li XS*, Sakurai M*, Kawamura A, Rose NR, Ng SS, Quinn AM, Rai G, Mott BT, Beswick P, Klose RJ, Oppermann U, Jadhav A, Heightman TD, Maloney DJ, Schofield CJ, Simeonov A. *PLoS One.* 2010; **5**(11):e15535, DOI: [10.1371/journal.pone.0015535](https://doi.org/10.1371/journal.pone.0015535)

Development of homogeneous luminescence assays for histone demethylase catalysis and binding.

Kawamura A, Tumber A, Rose NR, King ON, Daniel M, Oppermann U, Heightman TD, Schofield C. *Anal Biochem.* 2010; **404**(1):86-93, DOI: [10.1016/j.ab.2010.04.030](https://doi.org/10.1016/j.ab.2010.04.030)

Selective inhibitors of the JMJD2 histone demethylases: combined nondenaturing mass spectrometric screening and crystallographic approaches.

Rose NR, Woon EC, Kingham GL, King ON, Mecnović J, Clifton IJ, Ng SS, Talib-Hardy J, Oppermann U, McDonough MA, Schofield CJ. *J Med Chem.* 2010; **53**(4):1810-8, DOI: [10.1021/jm901680b](https://doi.org/10.1021/jm901680b)

Large Scale Structural Analysis of the Classical Human Protein Tyrosine Phosphatome.

Barr AJ, Ugochukwu E, Lee WH, King ON, Filippakopoulos P, Alfano I, Savitsky P, Burgess-Brown NA, Müller S, Knapp S. *Cell.* 2009; **136**(2):352-63, DOI: [10.1016/j.cell.2008.11.038](https://doi.org/10.1016/j.cell.2008.11.038)

* equal contributions

Contributions to macromolecular structures in the RCSB Protein Data Bank (with HUGO name)

Oxygenases:

[3NJY](#), [3PDQ](#), [3RVH](#), [4GD4](#), [5LY1](#) (KDM4A), [3OD4](#), [4AI8](#) (HIF1AN)

Phosphatases:

[3B7O](#) (PTPN11), [2QEP](#) (PTPRN2), [2P6X](#) (PTPN22), [2JJD](#) (PTPRE)

Kinases:

[3CEK](#), [3GFW](#), [3H9F](#) (TTK), [2VAC](#) (PTK9L), [2VUW](#) (GSG2A), [2VZ6](#) (CAMK2A), [2VX3](#) (DYRK1A), [2WNT](#) (RPS6KA1), [2WU6](#), [2WU7](#), [3RAW](#) (CLK3), [2WEI](#) (*Cryptosporidium parvum* CDPK1), [3LL6](#), [4O38](#) (GAK), [3OP5](#) (VRK1)

Other:

[3BPT](#) (HIBCH), [3DA2](#) (CA13), [2W2J](#) (CA8), [2V1X](#) (RECQL)